

Topic: General Fourier Series

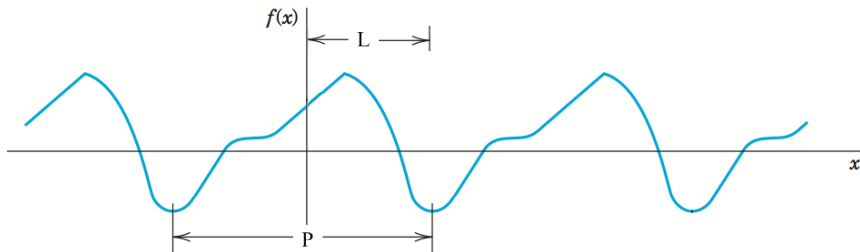
I. Definition

Defn: Suppose that $f(x)$ is periodic with period $P = 2L$. The **Fourier series** of f is

$$f(x) = \sum_{n=-\infty}^{\infty} d_n e^{i\pi n x / L}$$

where d_n are the complex **Fourier coefficients** of $f(x)$ given by

$$d_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-i\pi n x / L} dx$$



II. Properties of Even and Odd Functions

Type	Definition	Graphical Symmetry	Integral	Example
$g(x)$ is even			$\int_{-L}^L g(x) dx = 2 \int_0^L g(x) dx$	$\cos x$
$g(x)$ is odd			$\int_{-L}^L g(x) dx = 0$	$\sin x$

Product of odd & even functions			
$f(x)$	$g(x)$	$f(x)g(x)$	<i>Proof:</i> Let $h(x) = f(x)g(x)$
even	even	even	$h(-x) = f(-x)g(-x) = (f(x))(g(x)) = f(x)g(x) = h(x)$
even	odd	odd	$h(-x) = f(-x)g(-x) = (f(x))(-g(x)) = -f(x)g(x) = -h(x)$
odd	odd	even	$h(-x) = f(-x)g(-x) = (-f(x))(-g(x)) = f(x)g(x) = h(x)$

Ex: Identify each of the following functions as either odd, even, or neither

odd	even	neither
neither	odd	even
neither	odd	

III. Fourier Coefficients for Even and Odd Functions

Rewriting the Fourier Coefficients (d_n) using Euler's Formula ($e^{i\theta} = \cos \theta + i \sin \theta$) we observe

$$\begin{aligned}d_n &= \frac{1}{2L} \int_{-L}^L f(x) e^{-i\pi nx/L} dx = \frac{1}{2L} \int_{-L}^L f(x) \left[\cos\left(\frac{-\pi nx}{L}\right) + i \sin\left(\frac{-\pi nx}{L}\right) \right] dx \\&= \frac{1}{2L} \int_{-L}^L f(x) \cos\left(\frac{\pi nx}{L}\right) dx + \frac{-i}{2L} \int_{-L}^L f(x) \sin\left(\frac{\pi nx}{L}\right) dx\end{aligned}$$

Therefore, if

$$f(x) \text{ is even: } d_0 = \frac{1}{2L} \int_{-L}^L f(x) dx = \frac{1}{L} \int_0^L f(x) dx$$

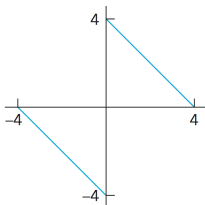
$$d_n = \frac{1}{2L} \int_{-L}^L f(x) \cos\left(\frac{\pi nx}{L}\right) dx + \frac{-i}{2L} \int_{-L}^L f(x) \sin\left(\frac{\pi nx}{L}\right) dx = \frac{1}{L} \int_0^L f(x) \cos\left(\frac{\pi nx}{L}\right) dx$$

$$f(x) \text{ is odd: } d_0 = \frac{1}{2L} \int_{-L}^L f(x) dx = 0$$

$$d_n = \frac{1}{2L} \int_{-L}^L f(x) \cos\left(\frac{\pi nx}{L}\right) dx + \frac{-i}{2L} \int_{-L}^L f(x) \sin\left(\frac{\pi nx}{L}\right) dx = \frac{-i}{L} \int_0^L f(x) \sin\left(\frac{\pi nx}{L}\right) dx$$

Summary of Fourier Coefficients when $f(x)$ is odd or even		
$f(x)$ is even	$d_0 = \frac{1}{L} \int_0^L f(x) dx$	$d_n = \frac{1}{L} \int_0^L f(x) \cos\left(\frac{\pi nx}{L}\right) dx$
$f(x)$ is odd	$d_0 = 0$	$d_n = \frac{-i}{L} \int_0^L f(x) \sin\left(\frac{\pi nx}{L}\right) dx$
$f(x)$ is neither	$d_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$	$d_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-i\pi nx/L} dx$

Ex: Find the Fourier Series of the given function



$$\cdot f(x) = \begin{cases} -x - 4 & -4 \leq x \leq 0 \\ -x + 4 & 0 \leq x \leq 4 \end{cases}$$

$$\cdot \text{Period} = 8 = 2L \implies L = 4$$

$$\cdot f(x) = f(x + 8)$$

$$\cdot f(x) \text{ is an odd function}$$

$$= 0$$

$$= \frac{-i}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx = \frac{-i}{4} \int_0^4 f(x) \sin\left(\frac{n\pi x}{4}\right) dx = \frac{-i}{4} \int_0^4 (-x + 4) \sin\left(\frac{n\pi x}{4}\right) dx$$

$$= \frac{-i}{4} \left[(-x + 4) \left(\frac{-4}{n\pi} \cos\left(\frac{n\pi x}{4}\right) \right) - \int \frac{4}{n\pi} \cos\left(\frac{n\pi x}{4}\right) dx \right]_0^4 \quad (\text{Integration By Parts : } u = -x + 4)$$

$$= \frac{-i}{4} \left[\frac{-4(-x + 4)}{n\pi} \cos\left(\frac{n\pi x}{4}\right) - \left(\frac{4}{n\pi}\right)^2 \sin\left(\frac{n\pi x}{4}\right) \right]_0^4$$

$$= \frac{-i}{4} \left[\left(\frac{-4(0)}{n\pi} \cos(n\pi) - \left(\frac{4}{n\pi}\right)^2 \sin(n\pi) \right) - \left(\frac{-4(4)}{n\pi} \cos(0) - \left(\frac{4}{n\pi}\right)^2 \sin(0) \right) \right]$$

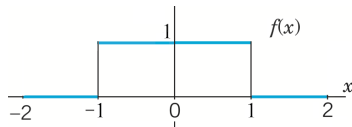
$$= \frac{-i}{4} \left[\frac{16}{n\pi} \right] = \frac{-4i}{n\pi}$$

$$\implies f(x) = \sum_{n=-\infty}^{\infty} d_n e^{in\pi x/L} \quad \text{where} \quad d_n = \begin{cases} 0 & n = 0 \\ \frac{-4i}{n\pi} & n \neq 0 \end{cases}$$

Ex: Find the Fourier Series of $f(x) = \begin{cases} 0 & -2 \leq x \leq -1 \\ 1 & -1 \leq x \leq 1 \\ 0 & 1 \leq x \leq 2 \end{cases}$ and $f(x) = f(x+4)$

Notes:

- Since $f(x) = f(x+4) \implies \text{Period} = 4 \implies L = 2$
- $f(x)$ is an **even** function



$$d_0 = \frac{1}{L} \int_0^L f(x) dx = \frac{1}{2} \int_0^2 f(x) dx = \frac{1}{2} \int_0^1 1 dx = \frac{1}{2}$$

$$d_n = \frac{1}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{1}{2} \int_0^2 f(x) \cos\left(\frac{n\pi x}{2}\right) dx$$

$$= \frac{1}{2} \int_0^1 1 \cos\left(\frac{n\pi x}{2}\right) dx = \frac{1}{2} \left[\frac{2}{\pi n} \sin\left(\frac{n\pi x}{2}\right) \right]_0^1 = \frac{1}{\pi n} \left(\sin\left(\frac{n\pi}{2}\right) - \sin(0) \right) = \frac{1}{\pi n} \sin\left(\frac{\pi}{2} n\right)$$

$$\implies f(x) = \sum_{n=-\infty}^{\infty} d_n e^{in\pi x/L} \quad \text{where} \quad d_n = \begin{cases} \frac{1}{2} & n = 0 \\ \frac{1}{\pi n} \sin\left(\frac{\pi}{2} n\right) & n \neq 0 \end{cases}$$

